

Summary

For non-collinear long range order (LRO) -

- Single **bond defect** acts as a **dipole**.
- With finite concentration of **bond defects**, LRO **survives** marginally.

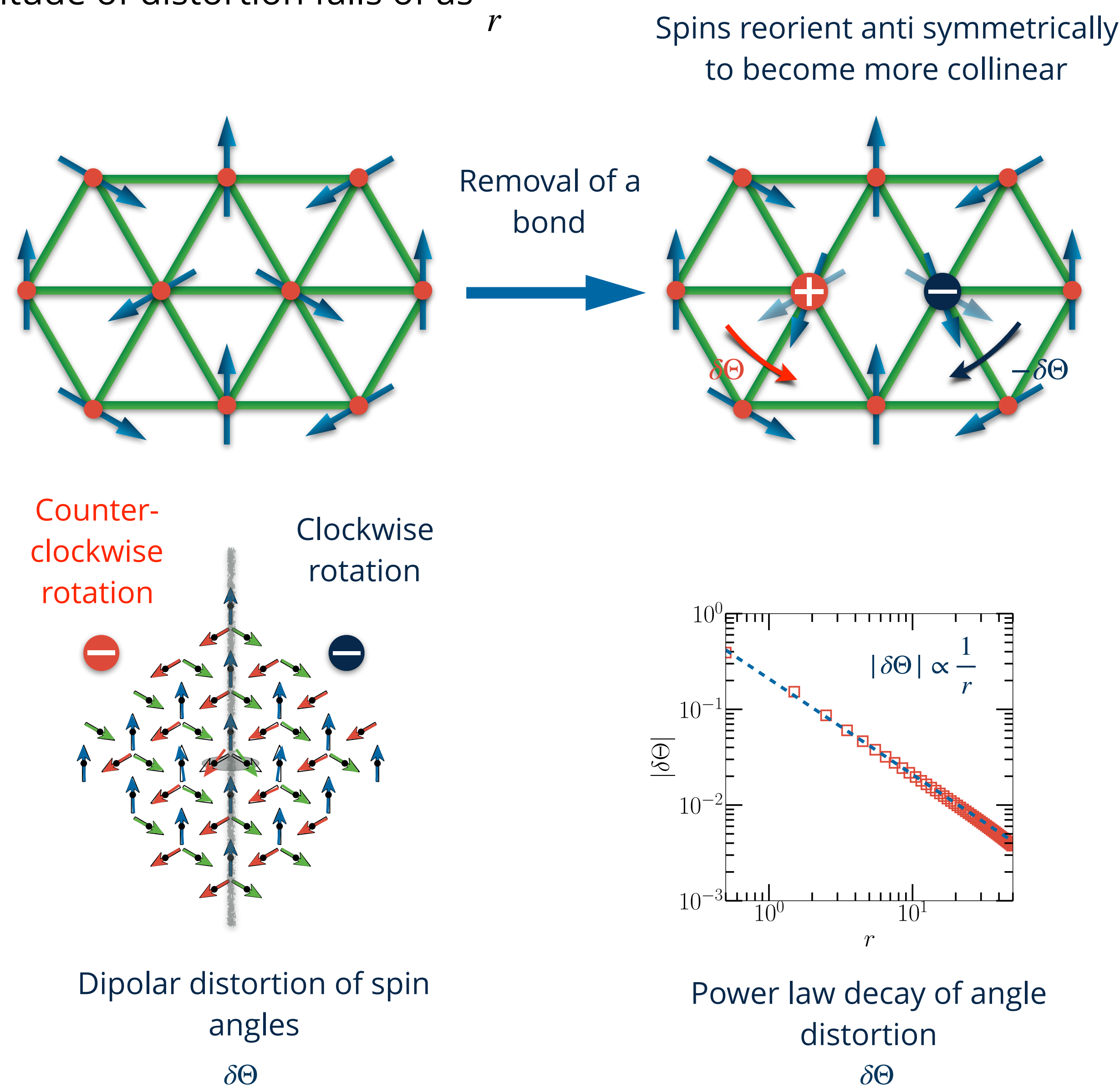
Single bond defect: emergent dipole

Triangular lattice (TLAF): Non-collinear Heisenberg magnet +120° spiral order. A single **bond defect** relieves frustration locally.

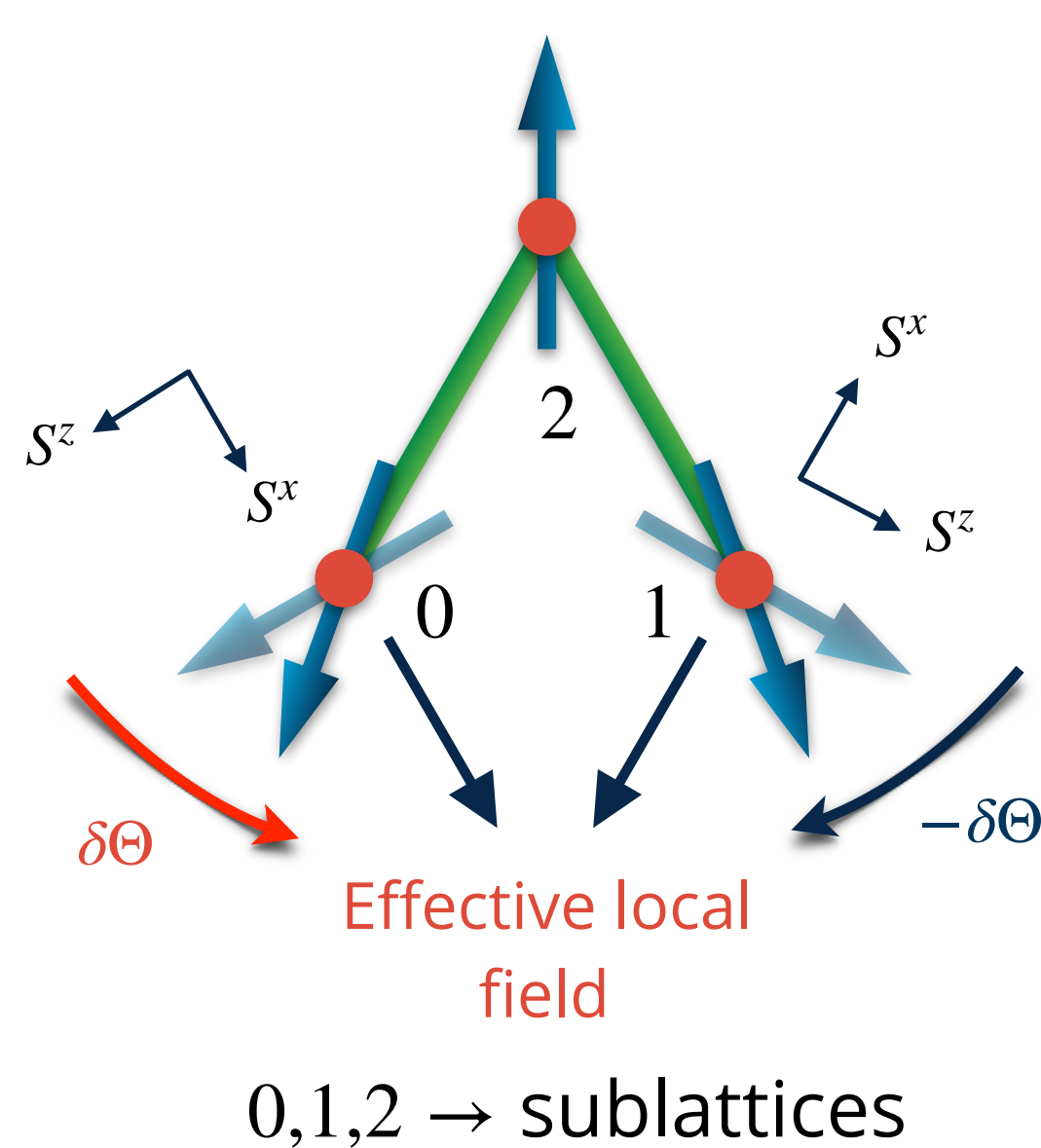
$$H = J \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + \delta J \mathbf{S}_0 \cdot \mathbf{S}_1$$

Bond defect leads to a **dipolar response**.

- Spins reorient **anti symmetrically** with dipolar mirror symmetry.
- Magnitude of distortion falls off as $\frac{1}{r}$



Local dipolar field in body frame:



Defect induced perturbation:

$$\delta H_i \sim \delta J \left(S_i^x - S_{i+a\mathbf{e}_{01}}^x \right)$$

with bond vector $a\mathbf{e}_{(01)}$.

In momentum space this leads to the **dipolar form factor** $(\mathbf{q} \cdot \mathbf{e}_{01})$.

$$\delta H_{\mathbf{q}} \sim \delta J (\mathbf{q} \cdot \mathbf{e}_{01}) S_{\mathbf{q}}^x$$

coupled to **in-plane goldstone modes**

$$\langle \langle S_{-\mathbf{q}}^x; S_{\mathbf{q}}^x \rangle \rangle \propto \frac{1}{q^2}$$

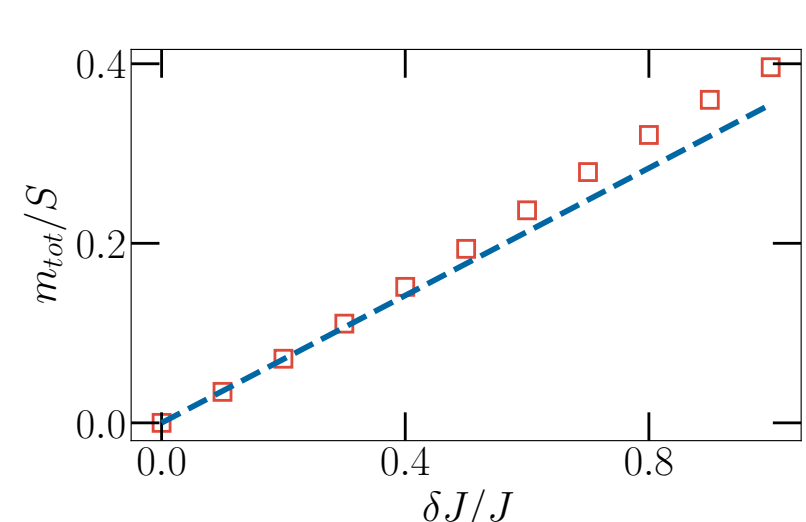
From the **linear-response** of the bond defect -

$$\delta\Theta(r) \propto \int d^d q e^{i\mathbf{q} \cdot \mathbf{r}} \frac{(\mathbf{q} \cdot \mathbf{e}_\alpha)}{q^2} \propto \frac{1}{r^{d-1}}$$

- In $d = 2$ this relates the **p-wave symmetry** of bond-defect with $\frac{1}{r}$ **dipolar decay**.
- Bond defect is a **stronger** effect compared to **vacancy** which has $\frac{1}{r^3}$ **octupolar** response [1].

Single bond defect: uniform moment

Spin-distortion generates a global uniform moment. In a $1/S$ expansion-



$$m_{tot} = (\delta J/J) S \left(0.39 - 0.08S^{-1} + \mathcal{O}(S^{-2}) \right)$$

Uniform moment - disorder dependence in TLAF

Bond-disordered non-collinear magnet: effective theory

- Bond defect $\delta J^{(\alpha\beta)} \rightarrow$ perturbation across bond direction $\mathbf{e}_{(\alpha\beta)}$
- Finite concentration of bond defects $\rightarrow$ Bond disorder.

Hamiltonian in each elementary plaquette-

$$H = J \sum_{\alpha \neq \beta} \mathbf{S}^\alpha \cdot \mathbf{S}^\beta + \sum_{\alpha \neq \beta} \delta J^{(\alpha\beta)} \mathbf{S}^\alpha \cdot \mathbf{S}^\beta$$

Goldstone modes generated on top of ordered state -

$$\mathbf{S}^\alpha(\mathbf{x}) = R(\mathbf{x}) \cdot \mathbf{S}_0^\alpha$$

Bond defect breaks the symmetry $R \rightarrow R^{-1}$, introduces **dipolar gradient**. Bond disorder leads to the coarse-grained quantum action -

$$S = - \int d\tau \int d^d x \text{Tr} \left[P R^{-1} \nabla_\mu^2 R \right] - \int d\tau \int d^d x \delta J^{(\alpha\beta)}(\mathbf{x}) e_{\alpha\beta}^i \text{Tr} \left[P T^3 (R^{-1} \nabla_i R) \right]$$

- For **triangular lattice**, $R \in SO(3) \simeq O(3)/Z_2$
- $\text{Tr} [P T^a T^b] = -(g/2) \delta^{ab}$ is stiffness of ordering with $T^a \in \text{Lie} [SO(3)]$ [2].
- T^3 , in-plane rotation generator, coupling associated **goldstone mode** to the **random dipolar field**.

Infrared catastrophe

Collection of bond defects with disorder averaging,

$$\overline{\delta J^{(\alpha\beta)}(\mathbf{x}) \delta J^{(\gamma\sigma)}(\mathbf{x}')} = \Delta^2 \delta^{(\alpha\beta);(\gamma\sigma)} \delta^{(d)}(\mathbf{x} - \mathbf{x}')$$

In $d = 2$, the **transverse** spin-correlation **diverges** -

$$\langle \overline{\mathbf{S}(\tau, \mathbf{x}) \cdot \mathbf{S}(\tau', \mathbf{x}')} \rangle \propto \Delta^2 \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')} \rightarrow \infty$$

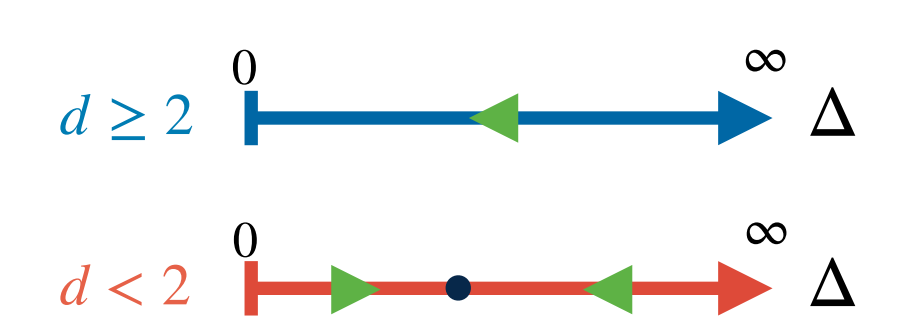
- For **non-vanishing disorder** strength $\Delta \rightarrow$ **breakdown** of LRO [3]!
- However, $[\Delta] = (2 - d)/2$ is **marginal** at tree level in $d = 2$. So higher-order perturbative corrections are needed to be considered.

Perturbative renormalization

With gaussian disorder distribution a **joint expansion** in $d - 2$ and Δ with the **replica method** -

Up to two-loop order :

$$\beta(\Delta^2) = (d - 2)\Delta^2 + \frac{\mathcal{N}}{g} \Delta^4$$

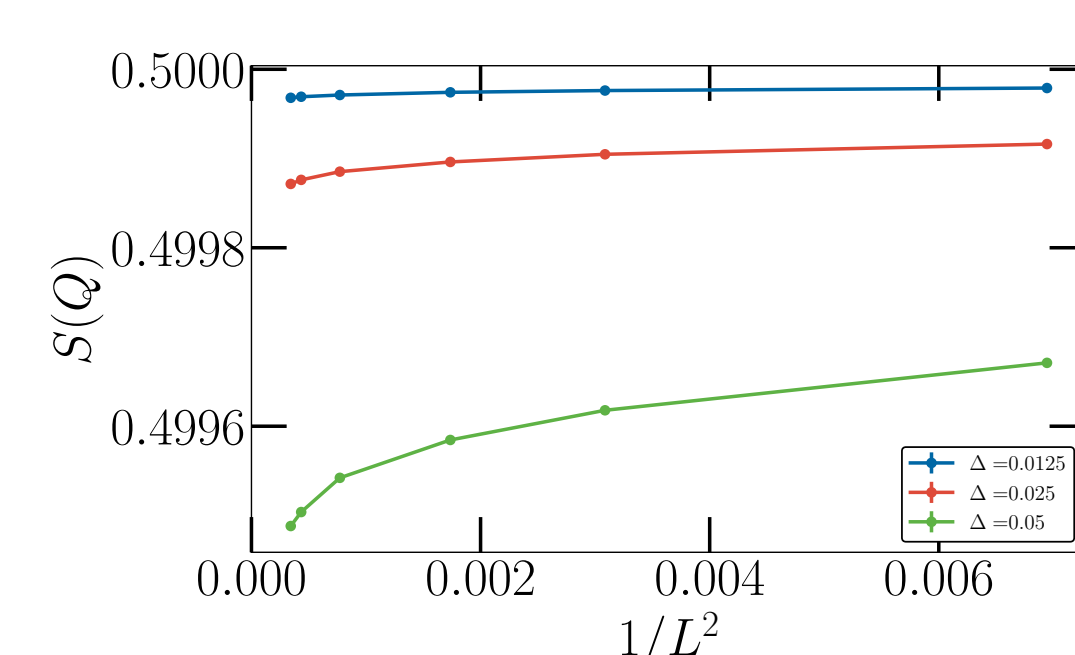


- Disorder is a **marginally irrelevant** perturbation in $d \geq 2$.
- So two-dimensional non-collinear LRO is **robust** against **weak disorder** [4].

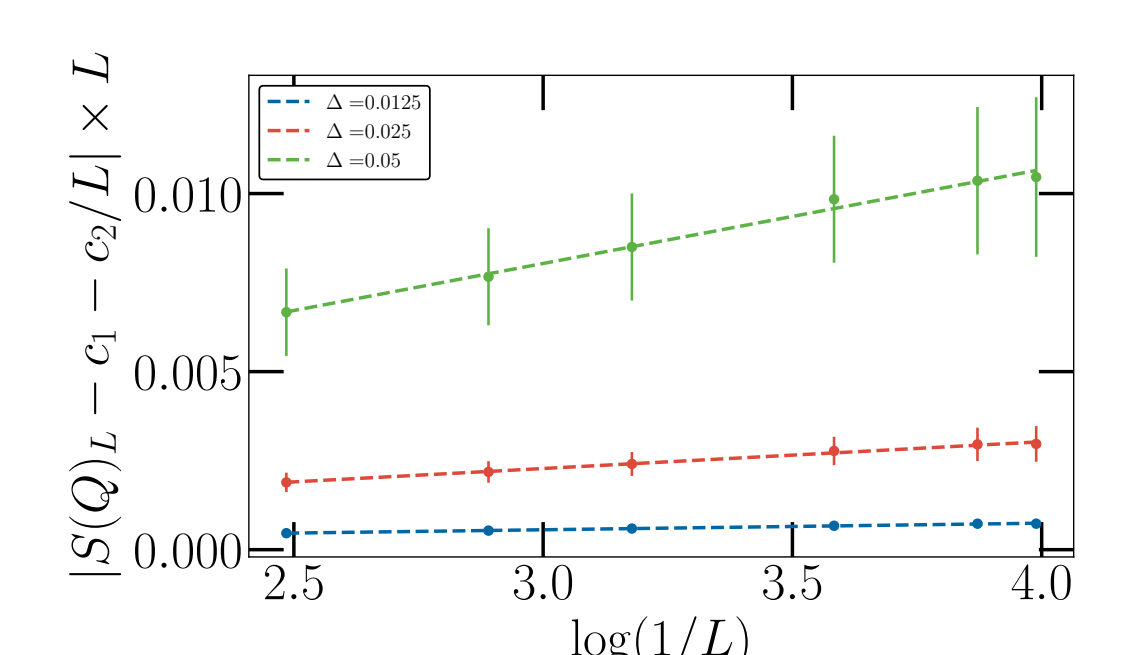
$$\dagger \mathcal{N} = \frac{1}{5!} \frac{(l-1)^2}{2\sqrt{2}\pi^3} \text{ with } l = \dim \left(\text{Lie} [SO(3)] \right)$$

Marginal disorder: numerical evidence (TLAF)

Classical triangular lattice Heisenberg AF with box distribution $\delta J/J \in [1 - \Delta, 1 + \Delta]$



Survival of LRO for weak disorder



Sub-leading scaling for marginal behavior

- c_1 = normalized Bragg peak intensity, c_2 = perturbative correction. An additional **log** correction from **marginal** perturbation is present.
- For classical spins LRO **survives** even for stronger disorder. But quantum fluctuations + strong disorder can **destroy** LRO [4,5].

References

- [1] A. Wollny, L. Fritz and M. Vojta, Phys. Rev. Lett. **107**, 137204 (2011)
- [2] P. Azaria et. al, Phys. Rev. Lett. **68** (1992) and Nucl. Phys. **B408**, 485 (1993), T. Dombre and N. Read, Phys. Rev. B **39**, 6797 (1989)
- [3] Y. Imry and S. Ma, Phys. Rev. Lett. **35**, 1399 (1975), A. Aharony, Solid State Comm. **28**, 667 (1978)
- [4] S. Dey and M. Vojta, in preparation
- [5] Wu. et. al, arXiv:1802.00935, Watanabe et. al, J. Phys. Soc. of Japan **83**, 034714 (2014)